

University of Victoria

Mathematical Analysis Project

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11.4 Periodic Points

For the section of periodic points, we're searching for the values in a math function which repeat themselves predictably. As long as you apply a function that takes a number and gives you another number. However, just in case there's a special point where, when you add a certain amount to it, you get back to the same function value, this is how a periodic point works.

In real analysis, periodic points are values in a mathematical function that exhibit a repeating pattern. To express in math language, a function S maps real numbers to real numbers, a point x is periodic if there exists a nonzero real number k . Once x is shifted by k units, the function's value is the same when at x and $x+k$. $S(x) = S(x+k)$

E. Give an example of a map with an orbit of period 6, but no odd orbits.

Suppose there exists a function map f in piecewise linear,

$$f(0)=1, f(1)=2, f(2)=3, f(3)=4, f(4)=5, f(5)=0$$

$$[0,1] \rightarrow [1,2] \rightarrow [2,3] \rightarrow [3,4] \rightarrow [4,5] \rightarrow [5,0] \rightarrow [0,1]$$

By the mapping process above, we can see the function has 6 cycles, every orbit has period 6. The function f has periodic points of even orbits but not odd orbits.

11.5 Chaotic Systems

From the topic of periodic points, we might have a bit idea about how a normal system starts and be predicted of where it goes. However, for chaotic systems, it's more like a super fancy roller coaster that runs unexpectedly every second.

In real analysis, once a small change has been made to a chaotic system at start, a huge difference in the outcome would be led to. It is trying to predict the exact future path a bunch of bouncing balls will take which might seem random, but it's actually following certain rules. The case is called a chaotic system which is dynamic and unpredictable mathematical systems that exhibit extreme sensitivity to initial conditions.

A. If T is topologically transitive on X , show that X either is infinite or consists of a single orbit.

To prove by contradiction, suppose there exists a topologically transitive system T on a space X is finite. Space X does not consist of a single orbit. There exist at least two distinct orbits which are O_1 and O_2 , and $O_1 \neq O_2$.

By $O_1 \neq O_2$, there exist distinct points $x \in O_1$ and $y \in O_2$, where $x \neq y$. If U is and V are both open neighborhoods, U is for x and V for y , they must have points from O_1 and O_2 .

The orbits are distinct and finite, there must exist a positive integer k such that

$$T^k O_1 \neq O_2, T^k O_2 \neq O_1.$$

$$T^k(U) \cap V = \emptyset, T^k(V) \cap U = \emptyset \text{ for all } n \geq 1.$$

From this we can see it is contradicting the assumption that the system is topologically transitive. It can't connect the open sets U and V through a finite sequence of applications of T . Therefore a topologically transitive system on a space X is either infinite or consists of a single orbit.

12.3 Differential Equations and Fixed Points

When you are giving a function that represents the position of an object, by applying a derivative it reaches its velocity, and then by applying the second derivative it reaches its acceleration. In real analysis, differential equations are the tools which guide us to how things change, also the rate at which these changes happen. A fixed point of that function is a special value where applying the function doesn't change anything. To express in math language, x is a fixed point because when you plug in x into the function, and back to x for function $f(x) = x$.

Overall, both differential equations and fixed points are crucial for dynamic studying. The differential equations help to understand how things change, and fixed points are the special values where things stay still from that change.

A. $y^{(3)} + y'' - x(y')^2 = e^x, \quad y(0) = 1, y'(0) = -1, \text{ and } y''(0) = 0.$

$$\int y^{(3)} dx + \int y'' dx = \int e^x dx + \int x(y')^2 dx$$

$$y'' + y' = e^x + 2 \int x(y') dx + 2$$

12.6 Linear Differential Equations

From the differential equations topic, we might have some idea of how something changes over time. For linear differential equations, it's when a differential equation is linear, the way things change is proportional to the current state. There is one important thing that the rate of change is directly tied to its current state. This is a method of capturing and predicting how things evolve over time in a clear, linear way.

A. Solve $y'' + 3y' - 10y = 8e^{3x}$, $y(0) = 3$, and $y'(0) = 0$.

$$r^2 + 3r - 10 = 0$$

$$(r+2)(r-5) = 0$$

$$r_1 = -2$$

$$r_2 = 5$$

$$y(x) = C_1 e^{2x} + C_2 e^{-5x} + e^{3x}$$

$$y_p = A e^{3x} = e^{3x}$$

$$y(0) = C_1 + C_2 + 1 = 3$$

$$y'(x) = 2C_1 e^{2x} - 5C_2 e^{-5x} + 3e^{3x}$$

$$y'(0) = 2C_1 - 5C_2 + 3 = 0$$

$$C_1 = 1 \quad C_2 = 1$$

$$y(x) = e^{2x} - e^{-5x} + e^{3x}$$